

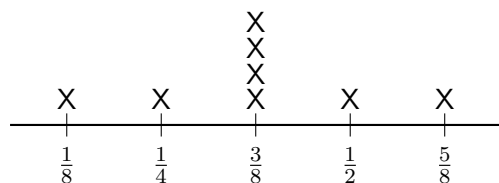
Reading Line Plots Marked in Fractions

Reading stacks of X's off a line plot marked in eighths and quarters, comparing the fractional values, and totalling or redistributing the whole set

How to use this sheet.

- Every X on a line plot stands for one measurement. The fraction written underneath says how big that measurement was.
- So a line plot holds two different kinds of number, and each question wants one of them. *How many* questions are answered by counting X's. *How much* questions are answered by adding the fractions the X's sit on.
- The marks along the bottom are evenly spaced, and equal spaces stand for equal differences in size. That is what lets you see the shape of the data at a glance.
- When you add fractions off a plot, rewrite them all with the same denominator first — eighths on the first plot, quarters on the second.

1. Count one stack. Ms. Oyelaran's class poured water into beakers and measured each one to the nearest eighth of a cup.



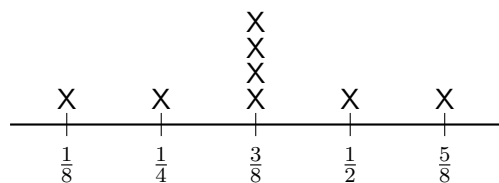
Water in each beaker (cups)

Each X stands for one beaker.

Find the mark for $\frac{1}{2}$ along the bottom, then count straight up.

How many beakers held exactly $\frac{1}{2}$ cup? _____

2. The tallest stack. Same beakers, same plot.



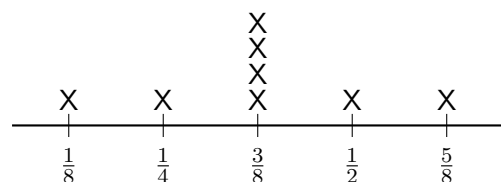
Water in each beaker (cups)

The tallest stack sits above the amount that came up most often.

Which amount was measured most often? _____

How many beakers held that amount? _____

3. How many more? Compare two stacks — not two fractions.



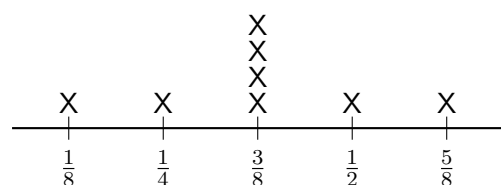
Water in each beaker (cups)

Both numbers you need are heights of stacks.

X's above $\frac{3}{8}$: _____ X's above $\frac{1}{4}$: _____

How many more beakers held $\frac{3}{8}$ cup than $\frac{1}{4}$ cup? _____

4. How many beakers? Every beaker put exactly one X on the plot.

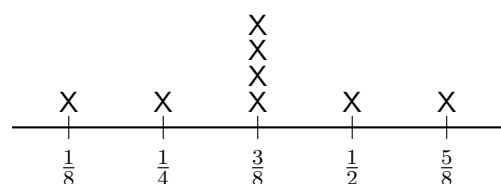


Water in each beaker (cups)

Count the X's, stack by stack, not the marks along the bottom.

How many beakers were measured in all? _____

5. How much water? This time the answer is an amount of water, so the fractions themselves have to be added.



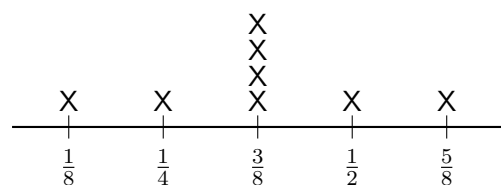
Water in each beaker (cups)

Write out one fraction for every X, then add. Eighths are easy to add once everything is written in eighths.

Rewrite $\frac{1}{4}$ as _____ eighths and $\frac{1}{2}$ as _____ eighths.

How much water is there altogether, in cups? _____

- 6. Share it out.** All the water is poured into one jug, then shared equally among the 8 beakers so that every beaker holds the same amount.



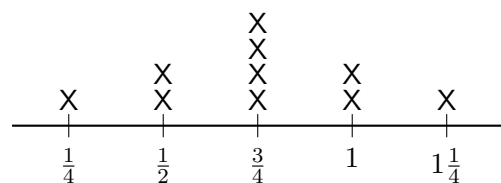
Water in each beaker (cups)

You already know the total from the problem before this one. Sharing equally is a division.

Total $\div$ number of beakers: _____ $\div$ _____

How much water would each beaker hold? _____

- 7. A new plot.** Ravi measured leaves to the nearest quarter of an inch. Each X is one leaf.



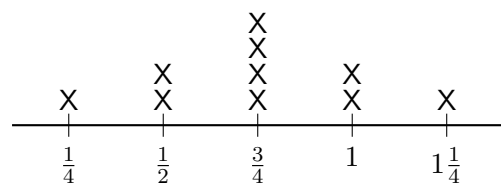
Leaf length (inches)

Read this plot the same way, but notice the marks now count by quarters.

How many leaves were $\frac{3}{4}$ inch long? _____

How many leaves were exactly one inch long?

- 8. Longest and shortest.** Find the X furthest to the right and the X furthest to the left.



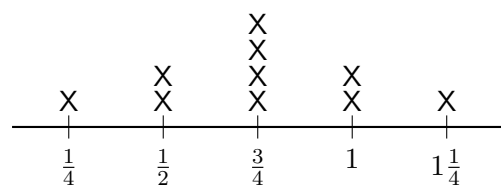
Leaf length (inches)

Longest leaf: _____ Shortest leaf:

How much longer was the longest leaf than the shortest one?

Answer: _____ in

9. Longer than three quarters. Put your pencil on the third mark from the left — the three-quarter-inch mark.



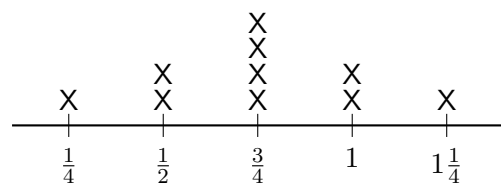
Leaf length (inches)

Every X to the *right* of that mark belongs to a leaf longer than three quarters of an inch.

X's on the next mark: _____ X's on the last mark: _____

How many leaves were longer than three quarters of an inch? _____

10. End to end. Ravi lays all his leaves in a line, end to end, with no gaps.



Leaf length (inches)

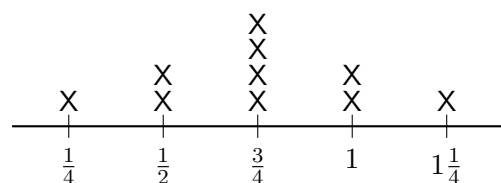
Add one length for every X. Writing every length in quarters first makes the adding easy.

Quarters altogether: _____

How long is the row of leaves? _____

Answer: _____ in

11. If every leaf were the same. Imagine the whole row of leaves could be cut up and shared out so that all ten leaves ended up exactly the same length.



Leaf length (inches)

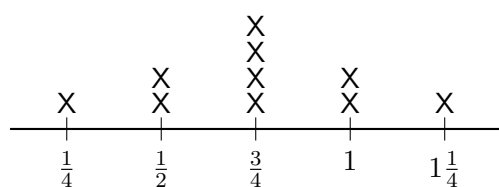
Same move as the beakers: total, then divide by how many.

Total $\div$ number of leaves: _____ $\div$ _____

How long would each leaf be? _____

Answer: _____ in

12. Two different questions. A line plot can answer “how long is one leaf?” and “how long are all the leaves together?” — and they are not the same question.



Leaf length (inches)

(a) How many leaves were $\frac{1}{2}$ inch long? _____

(b) Write one or two sentences saying which part of the plot answers each of the two questions above, and why counting X's can never answer the second one.
